

合肥训练场-线性度

1. 测试方案

在128节点和256节点分别进行dpsk万亿参数量模型的训练

对比两次训练的迭代耗时 计算线性度 计算公式为

$$(T1/T2) / 2$$
 要求结果 > 0.8

内部测试单双节点训练正常 loss收敛正常



参考示意图

2. 资源需求

合肥训练场的192个节点 训练测试过程中无占用 机器正常 无硬件故障 节点间通信正常

先在4节点上调试环境 k8s集群等 所需时间预估为8小时

调试正常后在千卡集群上进行训练任务

两次训练各3小时 观察loss收敛并记录单词迭代耗时

因千卡集群训练不可抗力因素较多 预留2小时部署环境 解决临时出现的报错 硬件故障等

3. 测试结果

3.1 环境准备

3.1.1 测试要求

万亿模型（本次使用deepseek-v3扩到1.1T参数规模）分别在128节点和256节点训练并记录平均单步迭代时间，计算集群线性度

3.1.2 conda环境

代码块

```
1 source /public/home/zhangyf9/miniconda3/etc/profile.d/conda.sh
2 conda activate ds_test
```

3.1.3 训练代码

代码块

```
1 cd /public/hgtest/deepseek_test/dcu_megatron/examples/deepseek_v3
2 bash run_deepseekv3_671B_128nodes.sh
```

3.1.4 配套

代码块

```
1 # 数据集
2 /public/source/train/deepseek/mmap_deepseekv3_datasets_text_document
3
4 # tokenizer
5 /public/hgtest/deepseek_test/DeepSeek-V3-671B
6
7 # dtk
8 /opt/dtk-25.04.2/env.sh
```

3.1.5 注意事项

- (1) 现场机器驱动版本较低，dtk版本最好不要超过25.04；
- (2) 谨慎修改env.sh文件；
- (3) 每次多机任务前建议先完成rccltest测试

3.2 训练实测

- 128节点 (gbs=2048)

```
[2026-07-18 22:57:19] iteration      2/ 286102 | consumed samples:      4096 | elapsed time per iteration (ms): 579
23.1 | throughput per GPU (TFLOP/s/GPU): 17.4 | learning rate: 3.993600E-07 | global batch size: 2048 | lm loss: 1.319043
E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 260.735 | number of skipped iterations: 0 | n
number of nan iterations: 0 |
[2026-07-18 22:58:14] iteration      3/ 286102 | consumed samples:      6144 | elapsed time per iteration (ms): 553
78.0 | throughput per GPU (TFLOP/s/GPU): 18.2 | learning rate: 4.040400E-07 | global batch size: 2048 | lm loss: 1.317833
E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 69.730 | number of skipped iterations: 0 | n
number of nan iterations: 0 |
[2026-07-18 22:59:10] iteration      4/ 286102 | consumed samples:      8192 | elapsed time per iteration (ms): 558
70.5 | throughput per GPU (TFLOP/s/GPU): 18.0 | learning rate: 4.087200E-07 | global batch size: 2048 | lm loss: 1.315159
E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 73.675 | number of skipped iterations: 0 | n
number of nan iterations: 0 |
[2026-07-18 23:00:05] iteration      5/ 286102 | consumed samples:     10240 | elapsed time per iteration (ms): 554
32.2 | throughput per GPU (TFLOP/s/GPU): 18.2 | learning rate: 4.134000E-07 | global batch size: 2048 | lm loss: 1.313613
E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 266.494 | number of skipped iterations: 0 | n
number of nan iterations: 0 |
[2026-07-18 23:01:01] iteration      6/ 286102 | consumed samples:     12288 | elapsed time per iteration (ms): 552
51.8 | throughput per GPU (TFLOP/s/GPU): 18.2 | learning rate: 4.180800E-07 | global batch size: 2048 | lm loss: 1.311477
E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 306.894 | number of skipped iterations: 0 | n
number of nan iterations: 0 |
```

- 256节点 gbs=4096

```
[2026-07-19 17:09:46] iteration     10/ 143051 | consumed samples:     40960 | elapsed time per iteration
(ms): 47593.7 | throughput per GPU (TFLOP/s/GPU): 21.1 | learning rate: 4.836000E-07 | global batch size: 4096 |
lm loss: 1.281482E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 70.899 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:10:36] iteration     11/ 143051 | consumed samples:     45056 | elapsed time per iteration
(ms): 49934.3 | throughput per GPU (TFLOP/s/GPU): 20.2 | learning rate: 4.929600E-07 | global batch size: 4096 |
lm loss: 1.277493E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 50.657 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:11:24] iteration     12/ 143051 | consumed samples:     49152 | elapsed time per iteration
(ms): 47609.9 | throughput per GPU (TFLOP/s/GPU): 21.1 | learning rate: 5.023200E-07 | global batch size: 4096 |
lm loss: 1.273437E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 10.970 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:12:10] iteration     13/ 143051 | consumed samples:     53248 | elapsed time per iteration
(ms): 46146.7 | throughput per GPU (TFLOP/s/GPU): 21.8 | learning rate: 5.116800E-07 | global batch size: 4096 |
lm loss: 1.261333E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 54.343 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:12:57] iteration     14/ 143051 | consumed samples:     57344 | elapsed time per iteration
(ms): 47531.1 | throughput per GPU (TFLOP/s/GPU): 21.2 | learning rate: 5.210400E-07 | global batch size: 4096 |
lm loss: 1.232972E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 121.552 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:13:45] iteration     15/ 143051 | consumed samples:     61440 | elapsed time per iteration
(ms): 47587.7 | throughput per GPU (TFLOP/s/GPU): 21.1 | learning rate: 5.304000E-07 | global batch size: 4096 |
lm loss: 1.225573E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 108.441 | number of
skipped iterations: 0 | number of nan iterations: 0 |
[2026-07-19 17:14:31] iteration     16/ 143051 | consumed samples:     65536 | elapsed time per iteration
(ms): 46169.1 | throughput per GPU (TFLOP/s/GPU): 21.8 | learning rate: 5.397600E-07 | global batch size: 4096 |
lm loss: 1.219219E+01 | seq_load_balancing_loss: 1.000402E+00 | loss scale: 1.0 | grad norm: 57.428 | number of
skipped iterations: 0 | number of nan iterations: 0 |
```

3.3 其他

3.3.1 deepseek-v3万亿扩展

修改下列参数

代码块

```
1  num-layers 61 --> 64
```

```
2 num-experts 256 --> 384
```

训练日志中搜索 `Total number of parameters in billions`，可以看到总参数量为 1049B：

```
Number of parameters in transformer block in billions: 1047.45
Number of parameters in embedding layers in billions: 1.85
Total number of parameters in billions: 1049.30
Number of parameters in most loaded shard in billions: 16.4822
Number of parameters in other shards in billions: 16.3664
```

3.3.2 deepseek-v4-pro 1.6T脚本备份

⚠ 该脚本为备份测试脚本，客户仅要求万亿模型，不要求具体模型，后续若有v4相关需求可使用

代码块

```
1 # dtk
2 source /public/hgtest/driver/dtk-26.04/env.sh
3
4 # 数据集
5 /public/hgtest/deepseek_test/oscar-dsv4/oscar-dsv4_text_document
6
7 # conda环境
8 source /public/hgtest/miniconda3/etc/profile.d/conda.sh
9 conda activate xunfei_dsv4
10
11 # 代码目录
12 cd /public/hgtest/deepseek_test/megatron-lm-dev-yh/examples/deepseekv4
13 bash run_deepseekv4.sh
```

3.3.3 常见问题

(1) NET/IB: Got completion from peer 10.121.129.24<41067> with status=5 opcode=0 len=0 vendor err 249 (Send) localGid ::ffff:10.121.12.55 remoteGids::ffff:10.121.18.24 hca mlx5_3

(2) 训练无错误异常退出，显示segfault

定位方法：

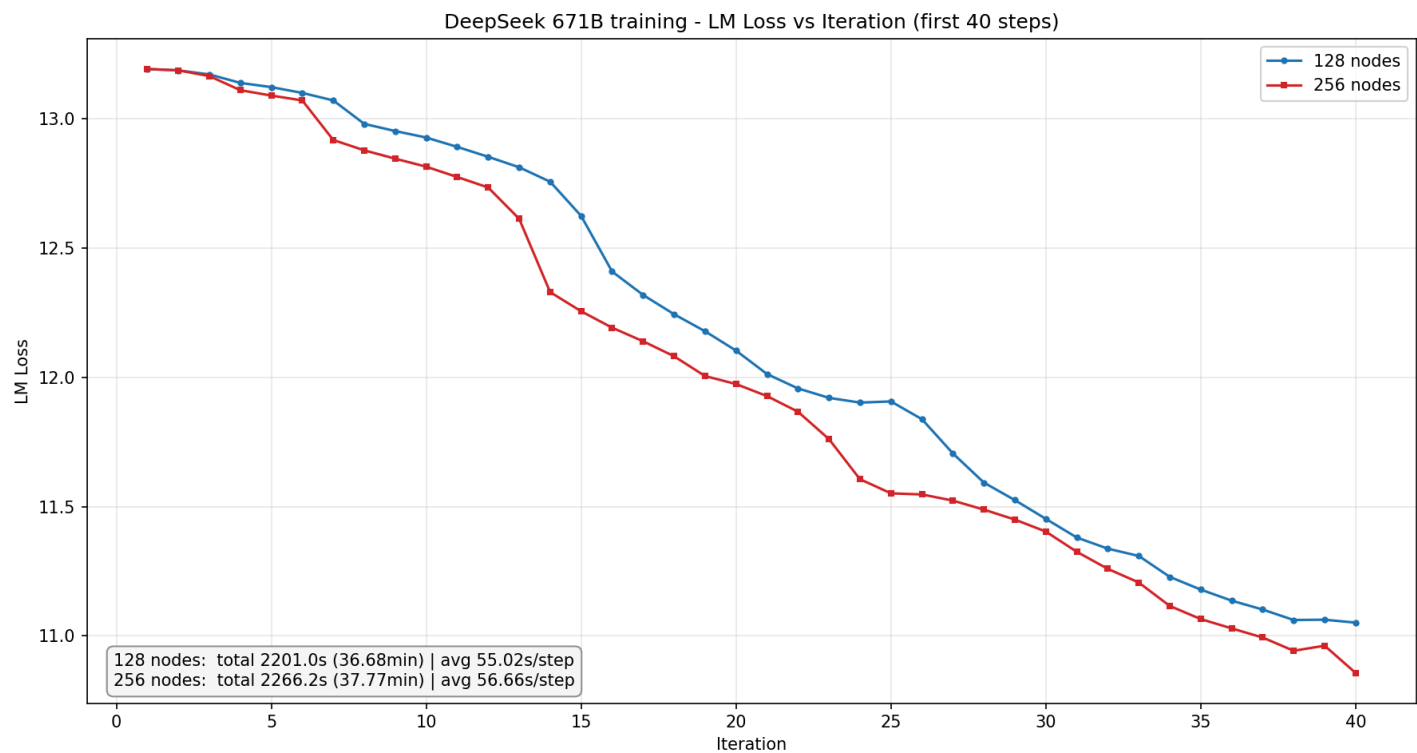
(1) 跑rccltest筛错误机器

(2) 检查如下项：

- `dmseg -T`
- 查看/var/log/syslog

- ipmitool sel list 命令查看输出
- 网络同事检查网络/网卡问题

3.4 结果



 xunfei_log-128nodes-2026-07-19-1212.log 
26.77MB

 xunfei_log-256nodes-2026-07-19-1653.log 
53.21MB

256集群训练的gbs（global batch size）是128集群的一倍 平摊到每张卡的mbs是一样的 算力翻倍 根据1-40步的总耗时计算线性度为97.1%